Digital Chip Design Program

RTL Design

Task #03

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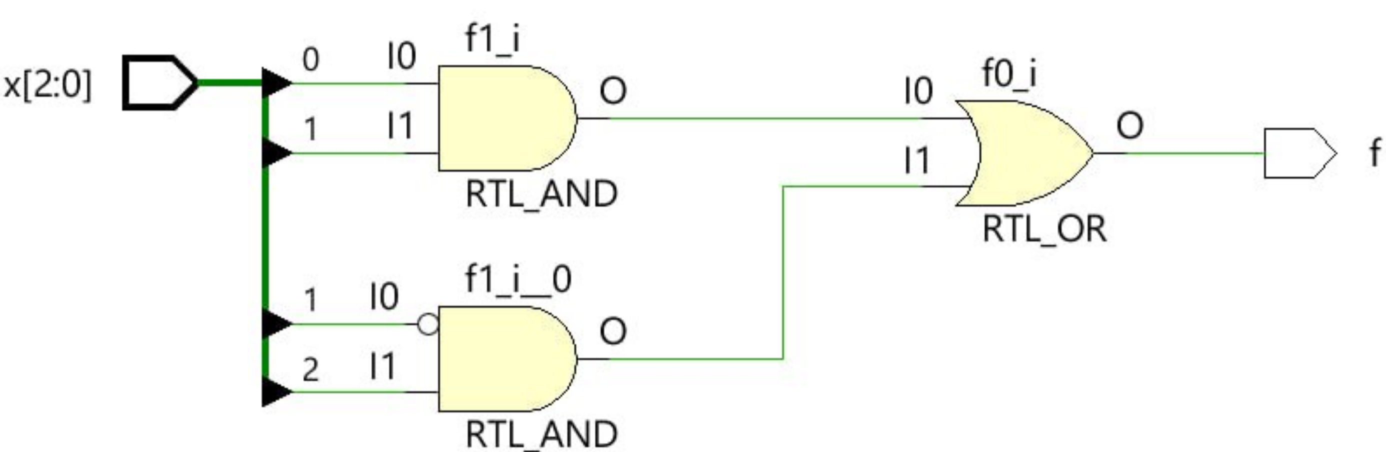
Behavioral

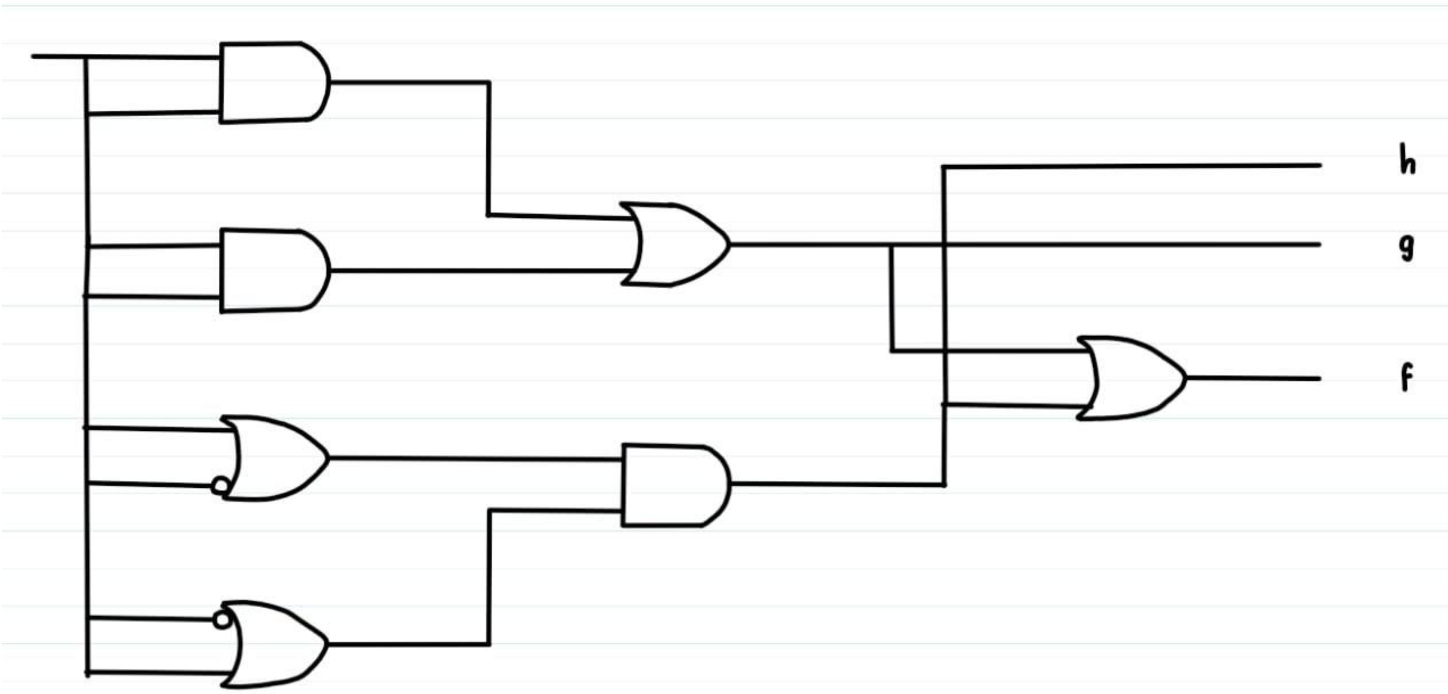
Version 1.0

Task #1.1

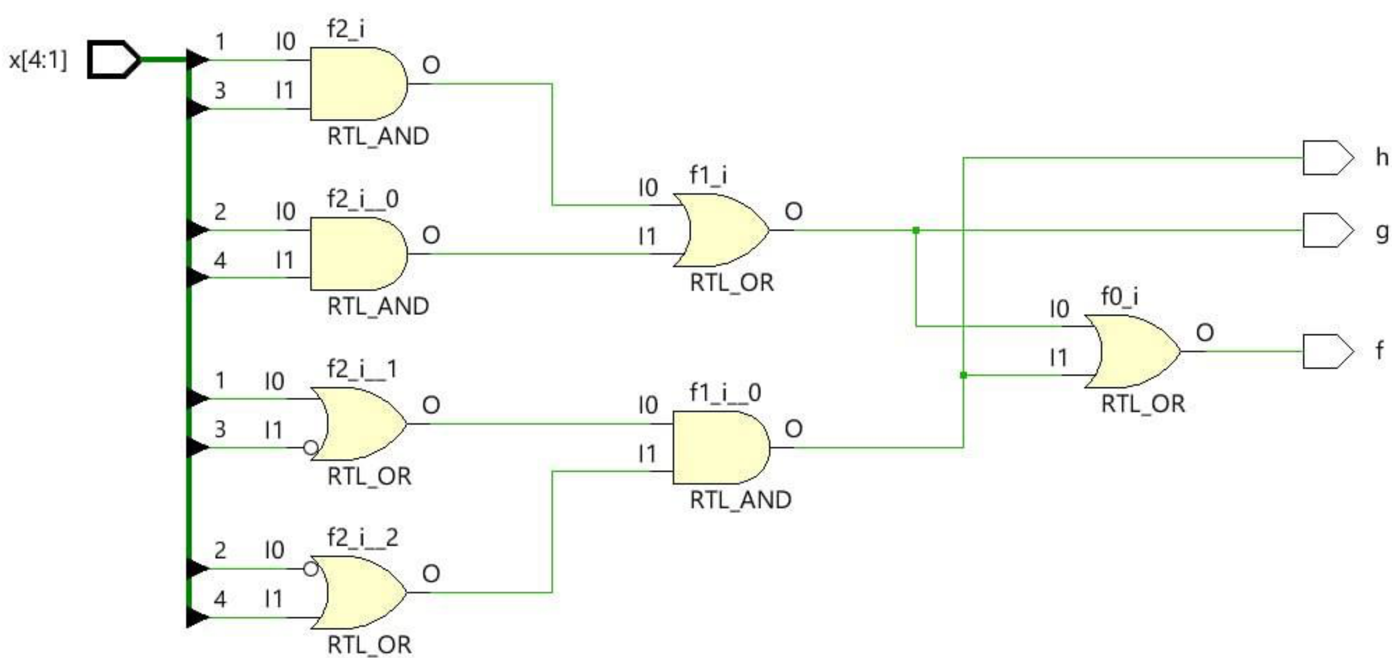
```
module task31
#(parameter n = 3) (
    input logic [n-1:0] x,
    output logic f
    );

    always_comb begin
        if ( (x[0] && x[1]) || (~x[1] && x[2]) ) begin
            f = 1;
            end
        else
            f = 0;
    end
endmodule
```



Task #1.2



h
g
f



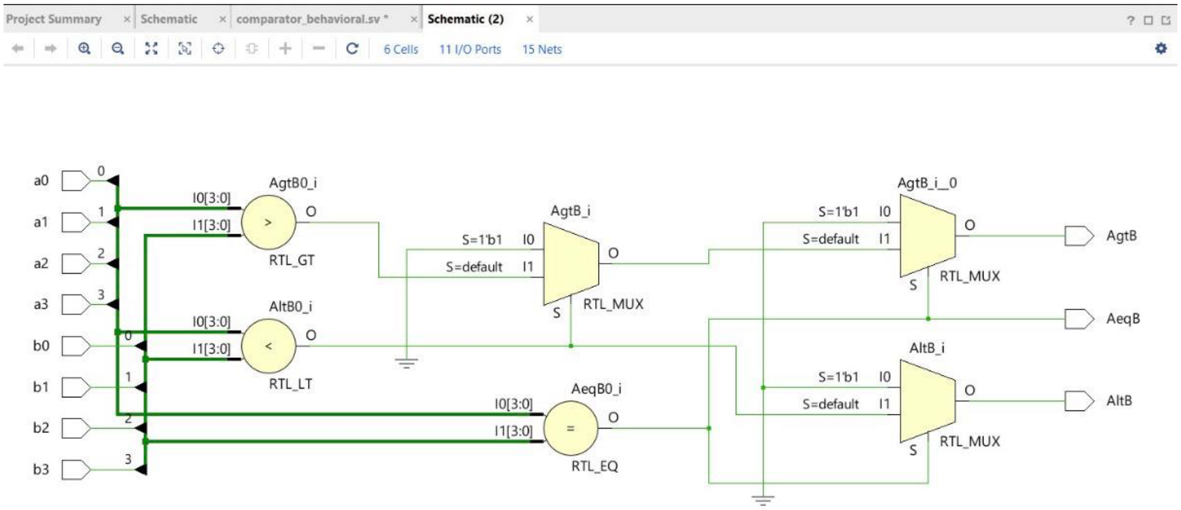
x[4:1]
1
3
I0
I1
f2_i
O
RTL_AND
2
4
I0
I1
f2_i__0
O
RTL_AND
1
3
I0
I1
f2_i__1
O
RTL_OR
2
4
I0
I1
f2_i__2
O
RTL_OR
I0
I1
f1_i
O
RTL_OR
I0
I1
f1_i__0
O
RTL_AND
I0
I1
f0_i
O
RTL_OR
h
g
f

Task #1.3

Project Summary × | Schematic × | comparator_behavioral.sv * × | Schematic (2) ×

C:/comparator_behavioral/comparator_behavioral.srcs/sources_1/new/comparator_behavioral.sv

```
module comparator_behavioral (
    input logic a3, a2, a1, a0,
    input logic b3, b2, b1, b0,
    output logic AeqB,
    output logic AltB,
    output logic AgtB
);
    logic [3:0] A;
    logic [3:0] B;
    always_comb begin
        A = {a3, a2, a1, a0};
        B = {b3, b2, b1, b0};
        // If A==B, then AeqB=1, AltB=0, AgtB=0
        if (A == B) begin
            AeqB = 1;
            AltB = 0;
            AgtB = 0;
        end
        // If A<B, then AeqB=0, AltB=1, AgtB=0
        else if (A < B) begin
            AeqB = 0;
            AltB = 1;
            AgtB = 0;
        end
        // If A>B, then AeqB=0, AltB=0, AgtB=1
        else if (A > B) begin
            AeqB = 0;
            AltB = 0;
            AgtB = 1;
        end
         else begin
            AeqB = 0;
            AltB = 0;
            AgtB = 0;
        end
    end

endmodule
```



Work distribution

Section	Farah&Joud	Tala&Horiah	Rana&Hoor&Sandra
Report	✓	✓	✓
Task 1.1			✓
Task 1.2	✓		
Task 1.3		✓	